C1 frozen policy diagnostic and reference review

PUBLIC revision

Scientific-only derivative of the research addendum

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This is a distinct scientific-only public revision. The frozen results and review judgments are unchanged. The public document does not provide full execution provenance. Public records include the selected-case classifications, event-response checks, scientific content hashes and generation settings; they are not sufficient to independently reconstruct or authenticate the full execution history. The empirical and review claims below must be read with this source-availability limitation.

Benchmark reference errors [R2], ambiguous questions and equivalent units limit what this diagnostic can establish. Its frozen parser-defined result is **11/64 questions (17.19%; Wilson 95% interval [R4] 9.88-28.21%)**. Assistant-assisted reconciliation corroborates **three numeric recovery witnesses on the same 64-question denominator**. This 3/64 count (4.69%) describes the support found under a conservative review rule; it does not replace the frozen endpoint or estimate a population recovery rate. The other eight parser events include three convention-dependent interpretations, four materially ambiguous cases and one match to an incorrect reference answer.

The study contains no training update. It does not measure useful reinforcement-learning gradients, reasoning improvement, controller superiority, held-out accuracy or general model capability. The prompts are public GSM8K training questions and are fresh only to this campaign. Historical V2 evidence remains separate and unpooled.

1 Frozen design and selection

C1 completed 4,096 fixed-order GSM8K [R1] screening questions with eight independently seeded responses per question. Eligibility required all eight outputs to be parseable, uncapped and assigned zero exact-match reward by frozen parser v2. After the full screen, the first 64 of 87 eligible questions received 24 additional responses each. Selection was truncated deterministically; the 23 other eligible questions were not followed up. The primary endpoint, a repeated-sampling coverage event [R3], is whether at least one of the 24 fresh responses receives parser-defined exact-match reward one. No question was removed or relabeled after review.

The new workload contains 32,768 screening responses and 1,536 fresh follow-up responses, totaling 34,304 new draws. A disjoint historical smoke cohort contributes 1,024 retained draws imported once, giving 35,328 recorded unique draws. The historical smoke is a separate quality-control cohort, not part of the primary denominator. The 512 initial responses for the selected 64 questions are already included within screening, and must not be counted again.

The model and tokenizer were Qwen/Qwen3-8B at revision `b968826d9c46dd6066d109eabc6255188de91218`, with BF16 and thinking disabled. The dataset was openai/gsm8k, main configuration, train split, revision `740312add88f781978c0658806c59bc2815b9866`. Generation used temperature 0.7, top-p 0.8, top-k 20, min-p 0, a 1,024-token generation cap and 4,096-token context. The recorded execution used one NVIDIA A100-SXM4-80GB, vLLM 0.11.0, PyTorch 2.8.0 and Transformers 4.57.1, with eager execution

enabled and prefix caching disabled. Fixed generation seeds do not imply bitwise reproducibility across hardware or execution environments.

The ordering seed was 20261002. Questions were ordered by ascending SHA-256 of the seed, the literal `order` and the SHA-256 of each whitespace-normalized question. The seed scheme used affine multiplier 15485863 modulo 2147483648 and generation-slot offset 262144. Follow-up slots use the original screening-cohort index, not the selected position. Exact initial and follow-up seed lists for all 64 selected cases are provided with the public method record [P2].

Screening had 504 parser failures and 19 cap hits among 32,768 outputs. By design the selected initial groups had neither. Follow-up had ten parser failures and four cap hits among 1,536 outputs. Those quality counts do not certify semantic correctness. The primary denominator remains 64 regardless of later quality flags. Descriptive prefixes of eight, sixteen and twenty-four fresh draws yield 9/64, 10/64 and 11/64 parser events. They are nested observations, not independent comparisons or evidence of an optimal rollout budget.

The Wilson interval uses a question-level binomial working model within the selected cohort. It is not design-based or population coverage: both outcome-dependent eligibility and fixed-order truncation restrict the estimand. The observed 87/4,096 eligibility count similarly describes this fixed screen, not benchmark-wide mathematical failure prevalence.

2 Review design and coverage

Specification gaps. The public method record [P2] gives the prompt suffix: "Solve the problem. Show your calculation briefly, then put the final numeric answer on its own line in the format: #### ." Thinking is disabled, but the complete rendered chat-template bytes are not provided here. The recorded ordering template is `seed:order:question_digest`; exact byte-level normalization conventions are not fully specified by that shorthand. Selected-case seed lists and relative affine strides are available, but the absolute seed-origin formula is not published and is not reconstructed from guessed constants. Reviewer model checkpoint identities and the selection rule for the 105 inspected non-success fresh responses are not recovered in this derivative. The 143 fresh responses therefore form a selected audit set, not a random validation sample. Parser-v2 code is identified by its hash in [P2], not fully specified by this prose. Independent human adjudication and release of audited raw texts remain outstanding; neither is represented as completed.

The first review solved the questions before comparing reference answers, then inspected the required model outputs. A second reference review was separately masked from the first judgments and model outputs. Final reconciliation compared the exact question, reference and annotations, retaining disagreements rather than forcing consensus. All stages were assistant-assisted. Separate masking and repeat review do not provide independent human validation or rule out shared model biases [R5].

The first output audit covers all 64 cases, all 512 initial responses and 143 required fresh responses: 655 responses in total, including all 38 primary-success outputs. The final mechanical coverage receipt verifies those identities and stored logic; it is not an independent mathematical adjudication. Six initially missing reasoning-caveat fields were clarified in a separately attributed supplement without altering the original fields, scores or verdicts. No coverage or annotation defect remains unresolved in the final receipt.

For the eleven event questions, reconciliation read all 88 initial outputs and all 38 primary-success fresh outputs in full. All 126 explicit final-number extractions are faithful and uncapped. Two unit-sensitive non-events add sixteen inspected initial outputs. This is not a claim to have validated reasoning in every one of the 1,536 fresh responses. The first audit identified five frozen-v2 extraction false negatives and no extraction false positives within its reviewed set; these selected-case observations do not establish population parser precision.

The conservative witness rule requires a reference value supported in both rounds without material ambiguity, eight faithful uncapped initial claims independently judged wrong, and at least one faithful uncapped fresh primary-success claim independently judged numerically correct. Sound final arithmetic and sound visible reasoning are recorded separately. No post-review filtering changes the 64-question denominator.

3 Frozen endpoint and conservative interpretation

Table 1: C1 results and interpretation. The final three rows partition the eight events outside the conservative witness count.

Quantity	Result	Interpretation
Eligible in the fixed screen	87 / 4,096	Eight parser-zero, valid and uncapped outputs
Selected for fresh draws	64 / 87	First 64 in frozen order
Frozen primary event	11 / 64	17.19%; Wilson 95% 9.88-28.21%
Conservative numeric witnesses	3 / 64	4.69%; support count, not a replacement endpoint
Convention-dependent event readings	3 cases	Retained review disagreement
Materially ambiguous events	4 cases	Requested quantity or premises unresolved
Wrong-reference event	1 case	Matching the reference reverses a correct initial answer

The three corroborated witnesses are Santana presents (C00, source rank 676), Suzanne bread (C07, rank 912), and Hank reading (C38, rank 2699). Santana's supported answer is $(4+7)-3 = 8$ presents; Suzanne's is $3\times2\times52/12 = 26$ loaves; Hank's is $5(30+60)+2\times2(30+60) = 810$ minutes. Each has eight reviewed, genuinely wrong initial final answers and at least one independently correct fresh final answer.

Santana also shows why final-answer correctness cannot validate reasoning. Its earliest successful fresh response, sample index 12, returns eight through an incorrect argument that no first-half birthday gifts were needed. Later successful indices 15 and 23 use the valid $11-3$ calculation. Hank's reference final value is supported even though its reference explanation mislabels subtotals. These distinctions remain visible in the event ledger. All sample and selected indices in this addendum are zero-based.

The checkpoint case (C35, rank 2593) gives the opposite result. Checkpoints at miles one and twenty-five enclose twenty-four miles across three gaps, so the separation is eight miles. All eight initial answers of eight are numerically correct. Four fresh outputs match the erroneous reference value six by dividing by four. The frozen analysis counts the event exactly as specified; semantic review does not call it mathematical recovery.

The citations, raft and fundraiser cases have favorable ordinary interpretations, but retain first-versus-second review disagreements. They are described individually rather than combined with the three corroborated witnesses into another adjusted recovery estimate. The remaining four cases concern aggregation of walking distance, the cake-and-ham probability event, the pet-owner denominator and Reggie's missing initial marble count. A gold match does not resolve those questions.

4 Reference quality and units

The original second-review categories are retained: sixteen clear wrong-reference judgments, nineteen materially ambiguous questions, two inconsistent questions, fifteen supported references and twelve references supported with conventional assumptions. Refinement separates four correct final answers with flawed reference explanations from the supported categories, leaving the following mutually exclusive view.

Table 2: Refined labels preserve the original classifications and distinguish final-value errors from explanation defects. They are second-review judgments, not a consensus benchmark error count.

Refined second-review category	Questions
Clear wrong reference final value	16
Materially ambiguous	19
Inconsistent question	2
Correct final value with flawed reference explanation	4
Supported	12
Supported with conventional assumptions	11
Total	64

A narrower cross-review analysis corroborates fifteen wrong-final-value judgments under the stated or ordinary mathematical model. The towel case explicitly assumes equal towel weights. Separately, two discrete-count questions have inconsistent premises and erroneous reference relations, three correct-final references have corroborated text defects, and two materially ambiguous questions contain explicit reference faults. These are **22 disjoint cases with specified reference defects, not 22 wrong labels**. The four correct-final cases in the refined second review and the three corroborated correct-final defects are different sets: a notation defect newly identified for Sonia/Joss is not corroborated in both rounds.

Thirteen first/second category disagreements are retained. The coin question illustrates the reason: gross deposits total 225 cents and the balance after withdrawal is 200 cents. The second review judges the recorded 200 wrong for the requested deposits, while the first retains a gross-versus-net target ambiguity.

It is excluded from the narrower fifteen. Ordinary mathematical conventions are not defects merely because every physical assumption is unstated; genuinely different quantities or plausible readings are described case by case. None of these audit counts estimates whole-GSM8K error prevalence because the cohort was selected for eight parser-zero outputs and then truncated.

Equivalent representations also matter. For Jackson ingredients (C36), all eight initial values 0.24 are equivalent to the reference 24 percent under the same conventional independent-selection probability interpretation; formal odds would be 6:19. The scalar parser extracted 0.24 faithfully. For Gunner plugs (C62), 200 pairs equal 400 individual plugs. Four initial responses preserve pair units, three remain unit-ambiguous and one incorrectly calls 200 the number of individual plugs. Neither case supports a claim of eight genuine mathematical errors. The frozen scalar references and rewards remain unchanged.

5 Separate prospective parser validation

The undeployed parser v3 candidate was frozen at 02:43:30 UTC on 3 October 2026, before the earliest target batch at 02:59:39 UTC. The fixed target was the last 256 screening questions, eight responses each, yielding 2,048 future outputs. This was a predefined future subset of an ongoing study, not preregistration before the entire C1 campaign. Historical candidate tuning is separate. C1's primary score remains based on frozen parser v2.

All 2,025 previously valid full parse objects were preserved. Four previously rejected outputs at four distinct prompts became accepted: one bold answer heading, two display-box blocks and one hash-prefixed XML answer block. Two separately blinded, output-only assistant annotations agreed on the same unambiguous scalar in all four changed cases. No changed acceptance was ambiguous or incorrectly extracted under those reviews. Nineteen outputs remained rejected.

This supports only a narrow pass on the four observed extraction changes. It does not establish semantic precision for the 2,025 unchanged outputs, validate the nineteen rejects, measure task correctness, or estimate population false-positive rates. The responses are clustered within prompts. Four examples do not exhaust their wrapper classes or malformed-input behavior. The candidate remains undeployed and is not used to recompute C1 recovery.

6 Conclusion

C1 reports a bounded frozen scoring result under the stated source-availability constraints and three conservative numerical counterexamples to the idea that eight observed wrong answers force all future answers to be wrong under this checkpoint, task and decoder. It also demonstrates why selection on exact-match zeros can enrich reference faults, ambiguity and representation mismatches. The unchanged 11/64 score and the 3/64 corroborated witness count answer different questions and must be reported together.

A next experimental step would require a new prospective design with independently reviewed references, explicit unit rules, random or fully enumerated follow-up selection, multiple seeds and a matched rollout budget. Any claim about training utility would additionally require an actual training intervention and held-out evaluation. Those experiments were not performed here. Historical V2 findings remain preserved as separate evidence.

7 Event ledger

The following ledger accounts for every frozen primary event. C labels use the zero-based selected index. Fresh indices identify all parser-success outputs among the 24 fresh draws. Scientific question, reference and response hashes and review notes accompany the public editable case ledger [P1].

Table 3: Complete accounting of all eleven parser-defined events.

Case and source rank	Reference	Fresh success indices	Interpretation
C00 Santana presents, 676	8	12, 15, 23	Corroborated
C07 Suzanne bread, 912	26	2, 5, 10, 15, 16, 21	Corroborated
C08 Citations, 923	4	0, 1, 3, 15, 23	Convention / disagreement
C19 Pet owners, 1482	100	4, 16	Ambiguous
C20 Raft capacity, 1544	17	23	Convention / disagreement
C29 Fundraiser, 2044	475	2, 4, 7, 13, 21, 22	Convention / disagreement
C35 Checkpoints, 2593	6	6, 7, 10, 22	Wrong reference
C37 Walking distance, 2660	36	0, 22	Ambiguous
C38 Hank reading, 2699	810	2	Corroborated
C47 Cake and ham, 3203	12	0, 4, 5, 9, 22	Ambiguous
C50 Reggie marbles, 3255	1	2, 15, 22	Ambiguous

C00 Santana presents. Gold 8 is (4 second-half birthdays+7 Christmas gifts)-3 March gifts. Both reviews support the quantity. Fresh 12 matches 8 through a false 8-0 argument; fresh 15 and 23 give the correct 11-3 calculation.

C07 Suzanne bread. 3 slices per meal $\times$ 2 weekend meals $\times$ 52 weeks/12 slices per loaf = 26. Sharing one slice does not require knowing the number of daughters. All 8 initial 90/91 claims use seven days per week; every fresh 26 uses the correct weekend schedule.

C08 Citations. Other citations most naturally means the combined non-parking count: L+L+2(L+L) = 24 gives 4. All initial 6 claims substitute twice one category. First treats that as material ambiguity; second supports 4. Ordinary-reading support is stronger than the per-category alternative, but disagreement remains and this is not added to the corroborated recovery count.

C19 Pet owners. Half of pet owners plus disjoint exhaustive dog/cat ownership gives 100; half of all citizens gives 300 under the same partition. Overlap allows other values. The denominator ambiguity is material even if ordinary classroom partition assumptions are granted.

C20 Raft capacity. The ordinary exact additive-space model makes 14 jackets occupy 7 person-spaces; 8 jackets reduce 21 by 4, yielding 17. Arbitrary real packing/safety possibilities do not by themselves constitute a defect. First conservatively withheld support; second accepts the conventional model. Retain that disagreement and conditional event support only in the ledger.

C29 Fundraiser. Next 4 days naturally names a single comparison period, making 950 the four-day total and shortfall 475 per student. Initial 0 answers insert per day before multiplying by 4; the text does not say each day. Retain the first alternative-scope objection and second conventional support, not a revised recovery estimate.

C35 Checkpoints. Interior route checkpoints at mile 1 and mile 25 enclose 24 miles and 3 intervening gaps, hence 8 miles. The closing 2-mile arc is outside those gaps. All 8 initials give 8; all 4 gold matches 6 divide by 4 and are wrong. Screen 1 has an internally contradictory parenthetical despite its correct final 8.

C37 Walking distance. Summed person-distance is $5 \times 3 \times 6+12+6 = 108$; shared outings counted once give 36. The aggregate quantity is genuinely different. Initial 0-6 are correct under person-mile reading; initial 7 incorrectly repeats weekly extras and gets 198. Neither gold-matching response proves the 108 answers wrong.

C38 Hank reading. $5 \times (30+60)+2 \times 2 \times (30+60) = 810$ minutes. Reference final sentence mislabels subtotals but final 810 is correct. Initial 630/690/1350 claims are wrong; fresh 2 visibly supplies the correct 450+360 calculation.

C47 Cake and ham. Cake-day overlap with any ham day during the week has probability 3/5 = 60%. A specified/randomly sampled test day has joint probability 3/25 = 12%. Both reviewer rounds retain sampling-unit ambiguity; all 5 fresh 12 matches use the latter event.

C50 Reggie marbles. With starting Reggie count R and losses L, $90 = R + 10 \times (9 - 2L)$, so $R = 20L$. His friend's 100 does not set R. Multiple feasible starts/losses exist. Gold's net-loss shortcut is invalid; all 3 fresh 1 matches have defective justifications. There is no uniquely correct loss count.

8 Complete reference classification index

This index preserves all 64 selected questions, including non-events. W = second-review wrong final value; A = material ambiguity; I = inconsistent premises; R = correct final value with flawed reference explanation; S = supported; C = supported with conventional assumptions. The category column uses the refined mutually exclusive mapping. An asterisk marks a retained first/second category disagreement. E marks a frozen parser event; N marks a conservative numeric witness. Neither marker changes the reference classification.

Table 4: Complete bounded-cohort index. The public JSON and CSV ledgers retain scientific case hashes, both review labels and case-specific explanations [P1].

Selected case	Source rank	Recorded gold	Category	Event and witness
C00	676	8	S	E N
C01	706	42	C	-
C02	754	200	W*	-
C03	803	200	A	-
C04	846	40	A	-
C05	855	625	S*	-
C06	878	480	W	-

Selected case	Source rank	Recorded gold	Category	Event and witness
C07	912	26	S	E N
C08	923	4	S*	E
C09	925	50	I	-
C10	945	1886	S	-
C11	950	41	W	-
C12	1078	2460	W	-
C13	1139	25	W	-
C14	1227	95	C*	-
C15	1266	49	W	-
C16	1307	301	S*	-
C17	1357	560	W	-
C18	1377	96	W	-
C19	1482	100	A	E
C20	1544	17	C*	E
C21	1739	750	A	-
C22	1743	50	W	-
C23	1780	3	A	-
C24	1872	35	S	-
C25	1897	2100	C*	-
C26	1921	23	W	-
C27	1932	140	A	-
C28	1996	99	R	-
C29	2044	475	C*	E
C30	2095	4	W	-
C31	2142	540	A	-
C32	2211	42	A	-
C33	2446	280	W	-
C34	2514	3	A*	-
C35	2593	6	W	E
C36	2617	24	C	-
C37	2660	36	A	E
C38	2699	810	R	E N
C39	2748	56	A	-
C40	2769	40	C	-
C41	2899	400	A	-
C42	2989	1100	A	-
C43	3021	9	C*	-
C44	3089	8000	A	-
C45	3118	350	R	-
C46	3119	163	S	-
C47	3203	12	A	E
C48	3209	1000	A	-

Selected case	Source rank	Recorded gold	Category	Event and witness
C49	3216	5	A*	-
C50	3255	1	A	E
C51	3259	983	I	-
C52	3277	84	C	-
C53	3299	128	W	-
C54	3335	30	W	-
C55	3409	7	R*	-
C56	3421	57	S	-
C57	3464	90	W	-
C58	3533	25	C*	-
C59	3536	6	S	-
C60	3579	40	S	-
C61	3581	4	A	-
C62	3593	400	S	-
C63	3597	52	C	-

9 Public scientific sources and availability

[P1] **Public selected-case ledger.** `case_ledger_public.json` preserves all 64 question texts, their scientific content hashes, reference answers and hashes, both reference-review labels, disagreements and reconciliation notes. It also preserves all eleven event records and 126 response checks, including response-content hashes, numeric verdicts and visible-reasoning caveats. `case_ledger_public.csv` is the compact classification view. The complete raw response corpus is not included in this derivative. A response hash alone does not allow its underlying response to be read or its judgment independently reproduced.

[P2] **Public method record.** `scientific_method_public.json` provides benchmark, model and tokenizer revisions, generation settings, ordering and seed parameters, and exact initial/follow-up seeds for the selected cases. These scientific parameters do not establish bitwise reproducibility across systems or authenticate execution.

[P3] **Public benchmark and model.** The study uses the GSM8K training split (`openai/gsm8k`, main configuration) and Qwen3-8B (`Qwen/Qwen3-8B`) at the revisions listed above.

Scholarly references. [R1] K. Cobbe et al., "Training verifiers to solve math word problems," arXiv:2110.14168, 2021 (GSM8K). [R2] J. Vendrow, E. Vendrow, S. Beery and A. Madry, "Do large language model benchmarks test reliability?," arXiv:2502.03461, 2025 (label errors in GSM8K-style benchmarks). [R3] B. Brown et al., "Large language monkeys: Scaling inference compute with repeated sampling," arXiv:2407.21787, 2024 (coverage under repeated sampling). [R4] L. D. Brown, T. T. Cai and A. DasGupta, "Interval estimation for a binomial proportion," *Statistical Science*, 16(2):101–133, 2001 (Wilson interval). [R5] L. Zheng et al., "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," arXiv:2306.05685, 2023 (limits of model-based adjudication, relevant to the assistant-assisted review stages).

This derivative reports the retained scientific analysis but is not a complete execution-provenance package or an independent replication. Original source labels have been replaced by the public scientific-source descriptions above. Hashes in the public ledger identify scientific content only; the public publication manifest binds the public files themselves. No hash supplies independent validation of mathematical correctness.